Internet Protocol Packet Structure and Addressing

Prof. Neminath Hubballi

Department of Computer Science and Engineering

Indian Institute of Technology Indore

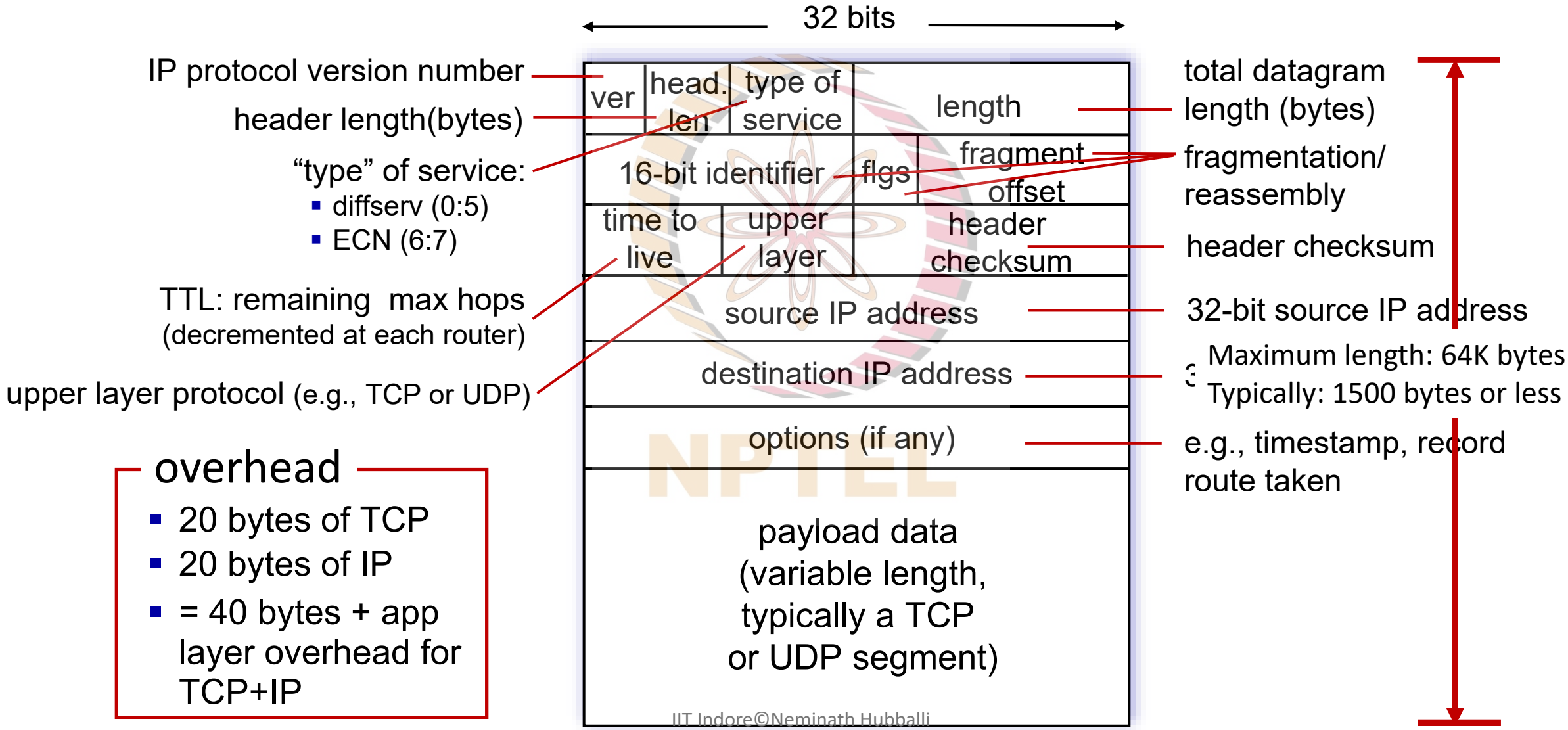
NPTEL

Overview

- IPv4 Header structure
- Network address
- Acquiring address
- Fragmentation
- Network Address Translation



IP Datagram Format

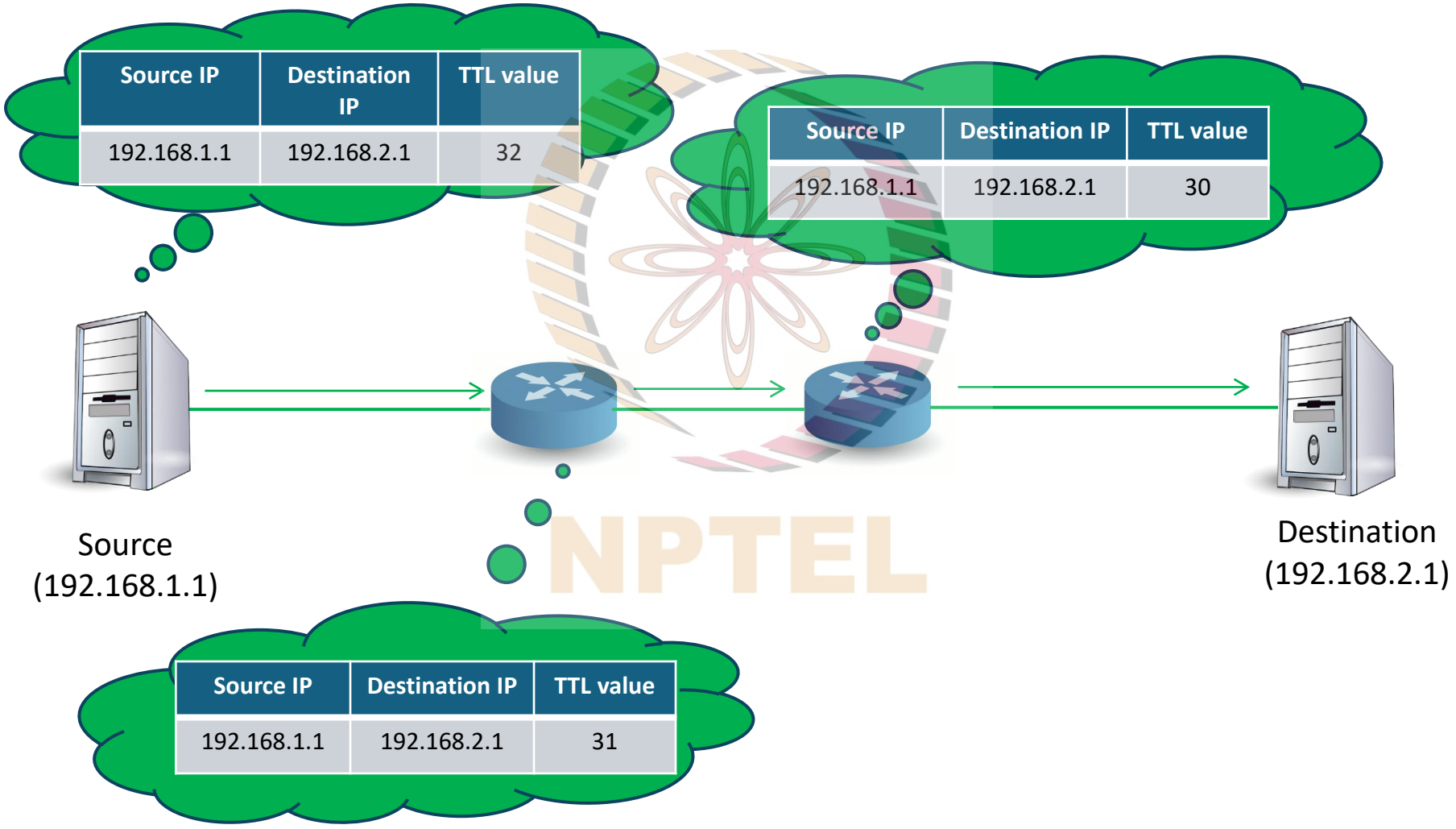


overhead

- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead for TCP+IP

IIT Indore©Neminath Hubballi

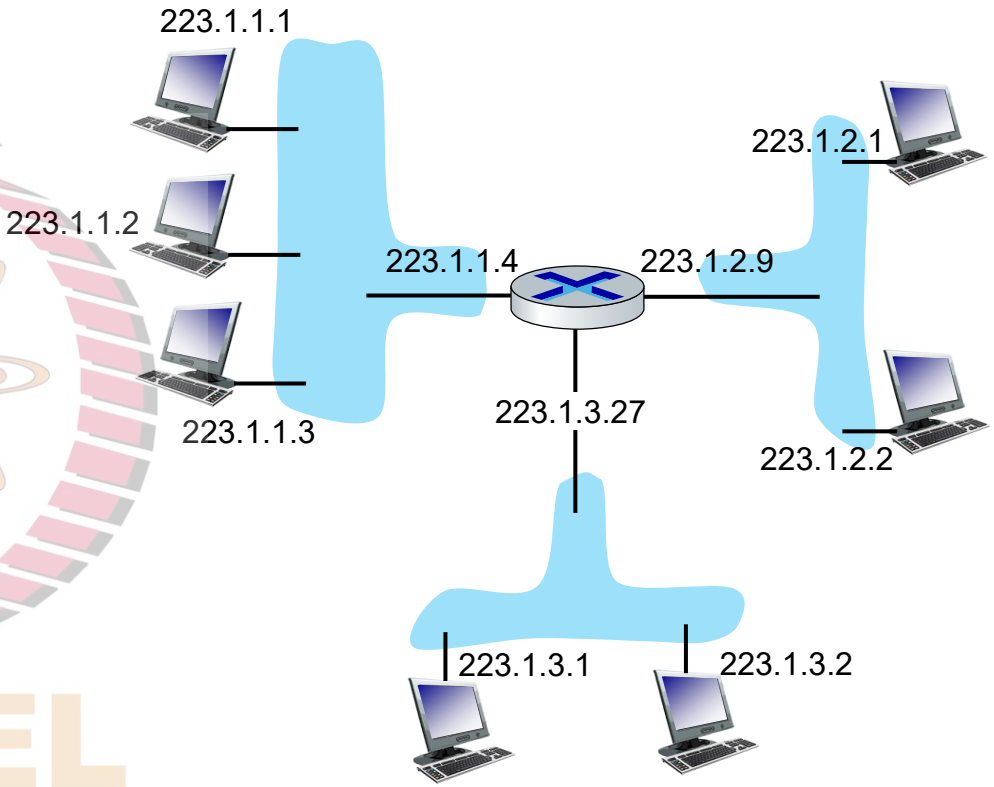
TTL Variation



Neminath Hubballi © Indian Institute of Technology Indore

IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **Interface:** connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

NPTEL

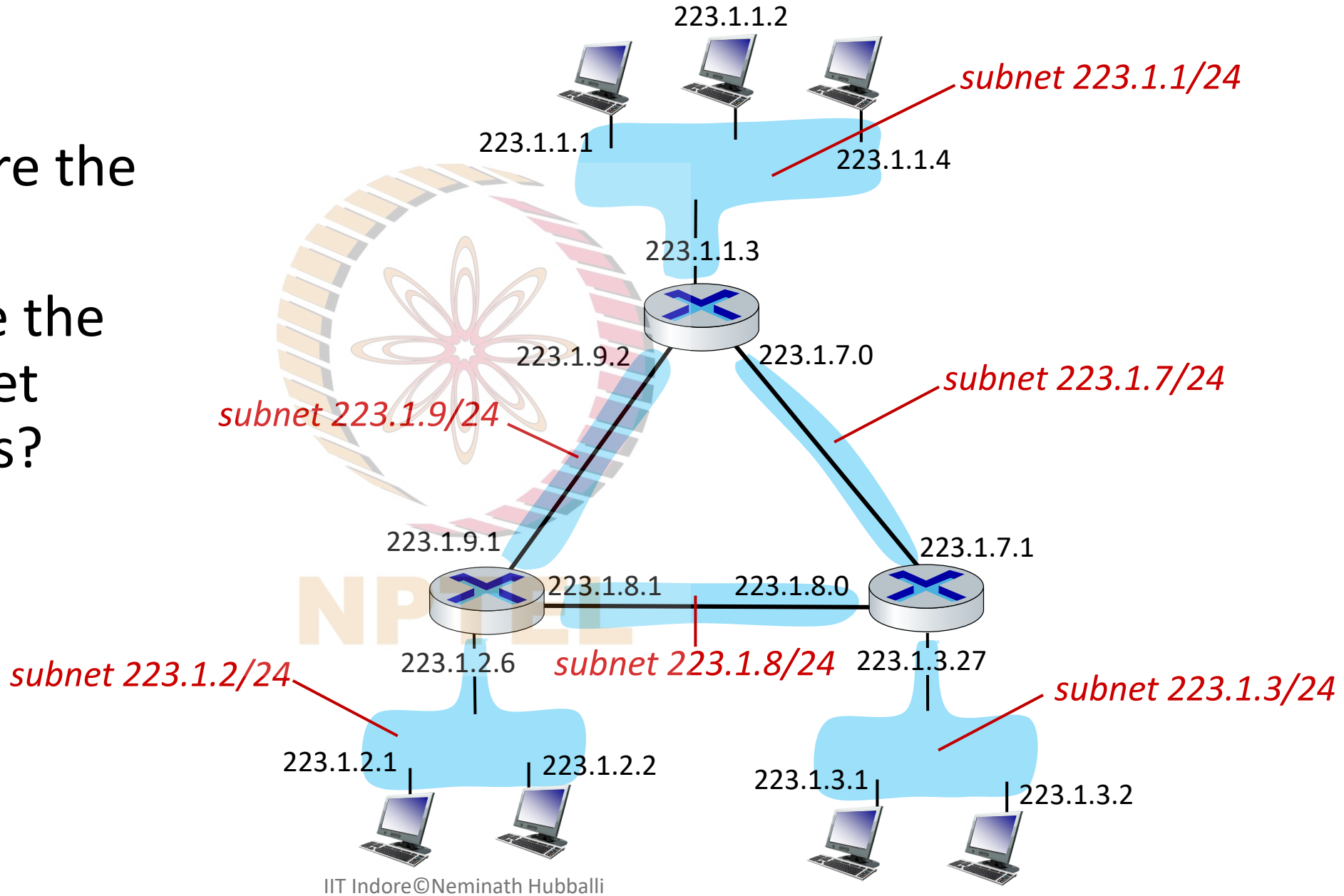
Subnet

- Subnet is a collection of device interfaces that can physically reach each other without passing through an intermediate router
- How to identify subnets ?
- IP addresses have structure:
  - subnet part: devices in same subnet have common high order bits
  - host part: remaining low order bits

NPTEL

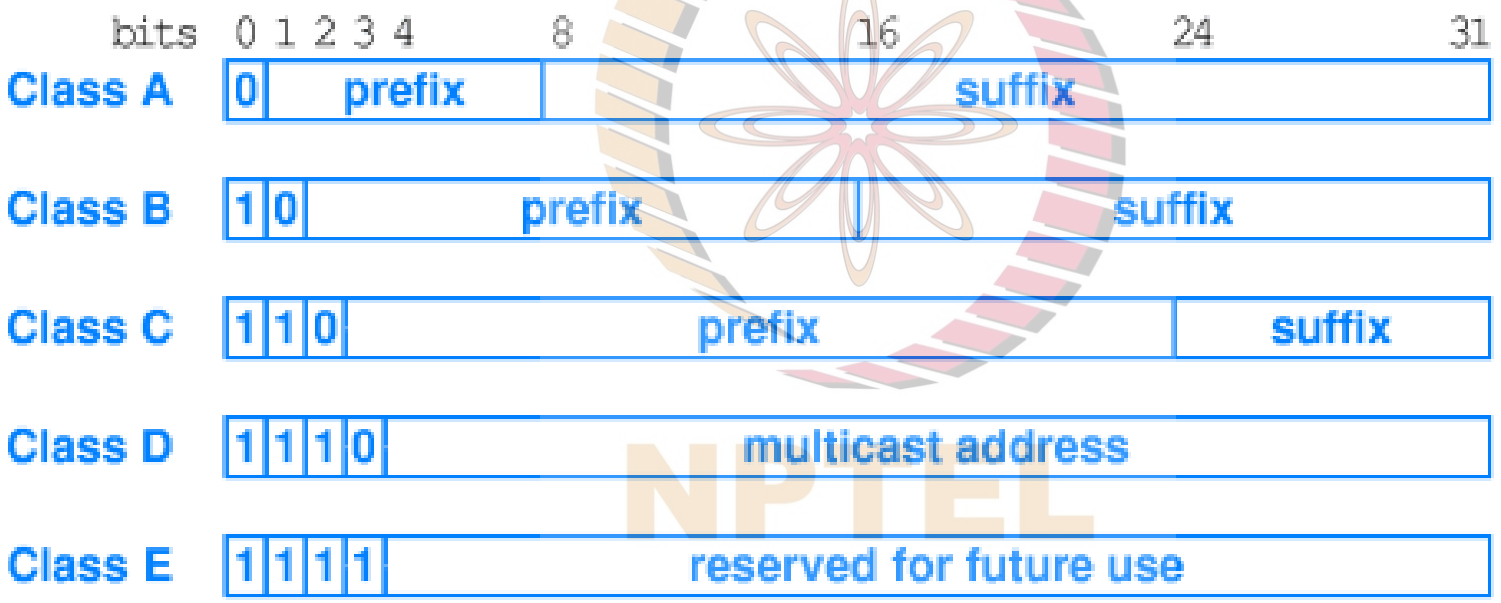
Subnets

- where are the subnets?
- what are the /24 subnet addresses?



IIT Indore©Neminath Hubballi

How to Create Subnets ?



NPTEL

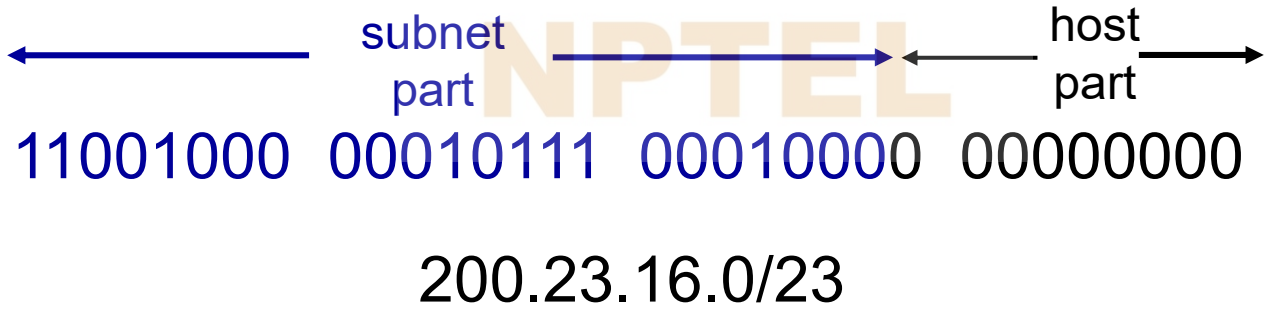
How to Create Subnets ?

Address Class	Bits In Prefix	Maximum Number of Networks	Bits In Suffix	Maximum Number Of Hosts Per Network
A	7	128	24	16777216
B	14	16384	16	65536
C	21	2097152	8	256

NPTEL

Variable Sized Networks

- Classless Inter Domain Routing (CIDR):
  - Subnet portion of address of arbitrary length
  - Address format: a.b.c.d/x, where x is # bits in subnet portion of address



200.23.16.0/23